
Understanding Grammar Representation in the Human Brain via Information-Restricted LLMs

Bachelor's Thesis

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Abstract

This thesis investigates the neural representation of grammatical class (GC) in the human brain, focusing on nouns and verbs. Using voxel-wise encoding models (VEM) and contextual embeddings from GPT-2, I examine whether the brain processes nouns and verbs in distinct areas. By applying attention masks to generate noun-masked and verb-masked embeddings, I fit these features to fMRI data collected from subjects listening to and reading naturalistic stories. The results reveal a complex pattern of GC representation that challenges traditional dichotomies and supports a more distributed account of language processing in the brain.

Zusammenfassung

Diese Arbeit untersucht die neuronale Repräsentation von grammatischer Klasse (GC) im menschlichen Gehirn, mit Fokus auf Nomen und Verben. Mithilfe von voxelweisen Enkodierungsmodellen (VEM) und kontextuellen Einbettungen von GPT-2 analysiere ich, ob das Gehirn Nomen und Verben in unterschiedlichen Bereichen verarbeitet. Durch die Anwendung von Aufmerksamkeitsmasken zur Generierung von Nomen- und Verb-maskierten Einbettungen passe ich diese Merkmale an fMRI-Daten an, die von Probanden gesammelt wurden, während sie natürliche Geschichten hörten und lasen. Die Ergebnisse zeigen ein komplexes Muster der GC-Repräsentation, das traditionelle Dichotomien herausfordert und eine stärker verteilte Darstellung der Sprachverarbeitung im Gehirn unterstützt.

Introduction

Language is a uniquely human capacity, and grammar is its organizing principle. Among the most fundamental categories of grammar are parts of speech (POS) — the distinctions between nouns, verbs, adjectives, and other word classes. In particular, the distinction between nouns and verbs has been a central focus of research in psycholinguistics and cognitive neuroscience. To investigate how the brain represents this distinction, researchers have employed various methodologies, including neuropsychological studies - which examine patients with brain damage - and brain imaging techniques, such as functional magnetic resonance imaging (fMRI).

Neuropsychological studies

Early neuropsychological studies conducted experiments with aphasic patients who had strong impairments in the retrieval and production of nouns and verbs respectively. A. R. Damasio and Tranel (1993), for example, observed impairments in noun retrieval for two patients with damage to the *left anterior and middle temporal lobe*, while the *left premotor cortex* was named as a possible cause of the third patient's impairment in the retrieval of verbs. This observation gave rise to the *fronto-temporal* dichotomy, suggesting that nouns might be processed in temporal areas while verbs are processed in frontal areas.

However, subsequent studies have found contradicting results, with some patients showing frontal damage without verb deficits and temporal damage without noun deficits (H. Damasio et al. 2001; De Renzi and Pellegrino 1995). Additionally, some of the verb deficits could be explained by the difference in imageability of the words (Crepaldi et al. 2013).

Brain Imaging

With the advancement of brain imaging techniques, such as fMRI, researchers have started to investigate the neural correlates of GC processing in healthy individuals. The results of these studies can be broadly categorized into three groups. First, some studies have found the neural distinction between nouns and verbs to be mainly driven by the semantic properties of the words, with nouns being more associated with objects and verbs with actions (Vigliocco et al. 2011). This view is often associated with the **emergentist** perspective in psycholinguistics, which posits that GC distinctions emerge from semantic constraints rather than being explicitly represented in the brain. Second, some studies have suggested that the difference can be attributed to grammatical operations like morphological inflection, which are more common in verbs than nouns. This view is often associated with the **combinatorial** perspective, which posits that GC distinctions are procedural and related to the combinatorial properties of words. Finally, some studies have found no clear distinction between the two classes (Crepaldi et al. 2013), pointing to a more **distributed** representation of GC in the brain.

The main limitation of these studies is that they rely on highly controlled stimuli, such as single words or simple sentences, which may not capture the complexity of natural language processing.

Naturalistic Studies

This limitation can be overcome by using Voxelwise Encoding Models (VEM). VEMs allow researchers to predict the BOLD response of individual voxels in the brain based on features extracted from naturalistic stimuli, such as stories or conversations. This approach has been used to investigate various aspects of language processing, including syntax, semantics, and stimulus modality (A. G. Huth et al. 2016; Deniz et al. 2019). Particularly with the advancement of large language models (LLMs) such as GPT-2 and their ability to generate contextualized embeddings, VEMs have become a powerful tool to investigate the neural representation of language in the brain.

I used VEMs to investigate the neural representation of nouns and verbs in the brain. After masking the attention of GPT-2 to either nouns or verbs during the generation of contextualized embeddings, I was able to create information-restricted feature-spaces. I then fit these feature spaces to the BOLD responses of 9 subjects who listened to and read naturalistic stories. Finally, I compared the performance of the noun-masked and the verb-masked model across the brain, to identify areas that were more sensitive to nouns and verbs respectively.

The central question of this thesis is whether the brain represents nouns and verbs in distinct areas, and if so, which of the three theoretical accounts — emergentist, combinatorial, or distributed — best characterises the observed pattern.

Materials and Methods

Stimuli

The speech stimuli consist of 11 short stories taken from **The Moth Radio Hour**, a storytelling program where true autobiographical stories are told in front of a live audience. These were used in previous papers (Huth et al., 2016), including the one where the response data was gathered (Deniz et al., 2019) providing a naturalistic stimulus.

Out of these 11 stories, 10 were used as training data while one 10-minute story was used as validation data. The latter was played two times during the experiment (in two separate scans) to which the responses were finally averaged across the two runs.

fMRI Data Collection and Preprocessing

Each spoken (and written) story was presented during a separate fMRI scan. The length of each scan was the same as the story. The sampling rate was 2 seconds, so $1 \text{ TR} = 2 \text{ s}$. Each scan included 10 seconds (5 TR) of silence both before and after the story. These data were collected during two 3-hour scan sessions that were performed on different days. MRI data were collected on a 3T Siemens TIM Trio scanner at the UC Berkeley Brain Imaging Center.

Each functional run was motion-corrected using the FMRIB Linear Image Registration Tool (FLIRT) from FSL 5.0 (Jenkinson and Smith, 2001; Jenkinson

et al., 2002). All volumes in the run were then averaged over time to obtain a high-quality template volume. FLIRT was also used to automatically align the template volume for each run with the overall template, which was chosen to be the temporal average of the first functional run for each participant.

Voxelwise Encoding Model Framework

Voxelwise encoding models (VEM) aim to predict the BOLD response for each individual voxel v from the stimulus S . This is usually done by first extracting one or more sets of features (called feature spaces) from the stimulus, then modeling the brain response as a linear combination of those feature spaces.

This corresponds to approximating the function $f(S) = g(S)\beta$ where g transforms the N -length stimulus into an $N \times P$ feature space with β as the $P \times V$ weights, V being the number of voxels. This can be accomplished by fitting the feature spaces $g(S_{\text{train}})$ to the responses R_{train} of a training dataset, yielding a model $\hat{f}$. The hemodynamic delay is accounted for by including delays of 0, 4, 8, and 16 seconds for each feature, resulting in a finite impulse response (FIR) model.

Finally, the model accuracy is calculated as the per-voxel Pearson’s correlation r_v between the model predictions on a validation dataset $\hat{R}_{\text{test}} = \hat{f}(S)$ and the actual responses R_{test} .

The value of r_v can then be interpreted as the extent to which voxel v ’s response to the stimulus can be explained by the given feature space. Projecting r_v onto the cortical surface then yields a sensitivity map to the specific feature space. This shifts the focus from experimental design to the choice of features, allowing researchers to test different hypotheses utilizing the same dataset by varying the feature space.

Banded Ridge Regression

To fit the multiple feature spaces to the BOLD responses I used Banded Ridge Regression (BRR), which is based on Ridge Regression, a linear regression method that augments the least-squares objective with an ℓ_2 penalty to reduce overfitting to the training data. This corresponds to finding the weight vector

$$\hat{\beta} = \arg \min_{\beta} \|X\beta - y\|_2^2 + \alpha \|\beta\|_2^2$$

with $\alpha \geq 0$ being the regularization parameter, which is typically selected through grid search or random search with cross-validation.

BRR extends Ridge Regression by assigning a separate regularization parameter α_i to each feature space X_i (Nunez-Elizalde et al. 2019).

$$\hat{\beta} = \arg \min_{\beta} \left\| \sum_{i=1}^m X_i \beta_i - y \right\|_2^2 + \sum_{i=1}^m \alpha_i \|\beta_i\|_2^2$$

This is important since the feature spaces can vary greatly in size and signal-to-noise ratio and thus need different levels of regularization.

Features

To reproduce the original study, the 985-dimensional Eng1000 features were provided.

To account for response variance caused by various modality-dependent nuisances, several additional features were provided.

Phonemes

A list of phoneme-time pairs was constructed for each of the 39 English phonemes. This list was then down-sampled to the fMRI acquisition rate. The result is a $(39 \times N_{TR})$ array. These features were used when fitting the model to the listening data.

Letters

Similar to the phonemes, a 26-parameter model was created to represent the frequency of each of the 26 letters in the English alphabet that appeared in the written stories. This was also down-sampled to the sampling rate, resulting in a $(26 \times N_{TR})$ array.

Additional Features

To account for the highly variable speech rate both within and across stories, single-feature models were constructed. These count the number of words, number of phonemes, number of letters, and number of story speaker’s pauses that occurred during the acquisition of each fMRI volume (2.0045 s). To account for the variable word lengths during the visual presentation, a single-feature word length variation model was constructed by taking the variance of word lengths that occurred during the acquisition of each fMRI volume.

Contextual Embeddings

The largest issue with the lexical *English1000* embeddings is the lack of contextual information in the feature space. Put differently, the same word will be mapped to the same feature vector regardless of the context it appears in. For example, the word *fine* has a very different meaning in the sentences “I am fine” and “I paid a fine”.

To account for this, I made use of *contextual embeddings*, which, as the name suggests, aim to include contextual information in the word embeddings. Such embeddings were introduced by the Long Short-Term Memory (LSTM) network ELMo (Peters et al. 2018) and refined by the transformer model BERT (Devlin et al. 2019).

Contextual embeddings play a fundamental role in all modern Language Models (LM), and for this thesis I decided to use GPT-2’s embeddings, as it is commonly used in related literature (Schrimpf et al. 2021; Tuckute et al. 2024).

Given an input sequence comprised of the i -th word of a story w_i and its k -length context $C_i = (w_s, \dots, w_{i-1})$ with $s = \max(0, i - k)$, I can generate the

contextual embeddings through the following steps.

Tokenization

Before a sentence or word can be embedded into vectors, it needs to be broken up into a set of token IDs. This turns the input sequence into a sequence of tokens $(t_1, \dots, t_L) \in \mathbb{N}^L$.

Initial Embeddings

Upon receiving the token sequence, GPT-2 first converts the tokens into embedding vectors via a *lookup table* with a fixed embedding size $d_{\text{model}} = 768$, then adds positional encodings to generate the initial embeddings. These initial embeddings are non-contextual.

Hidden Layers

The initial embeddings then go through a sequence of layers where they are first weighted using the attention mechanism (Vaswani et al. 2023), then passed through a feed-forward network to enrich them with contextual information from surrounding tokens. This yields a new contextualized embedding of the same dimensionality d_{model} .

Previous studies have found that the middle layers, e.g., Layer 8 for GPT-2 small, are best at predicting brain activity in VEMs (Schrimpf et al. 2021; Jain and A. Huth 2018), and Layer 8 is therefore used throughout this work.

Averaging

The hidden state $H_8 \in \mathbb{R}^{L \times 768}$ contains the contextualized embeddings for every token in the input sequence. These tokens are then averaged to return w 's contextualized embedding e_w :

$$e_w = \frac{1}{L} \sum_{i=1}^L h_8^{(i)}$$

Downsampling

After repeating this process for each word in each story, I have an embedding matrix $E_j \in \mathbb{R}^{N_j \times 768}$ with $N_j = |S_j|$ being the number of words in the j -th story. To fit the features to the fMRI data, the contextual embeddings are downsampled onto a feature space $F_j \in \mathbb{R}^{K_j \times 768}$ via a Lanczos filter, with K_j being the number of sample points (TRs) of the j -th story.

The baseline contextualized embeddings are generated using a context length of 20 words for each word in each story.

Context Masking

The masking of Parts of Speech (POS) was performed automatically using Python's `spaCy` library. This library first splits an input sequence into a series of word-like tokens based on language-specific rules. For example, *don't* is split into two tokens *do* and *n't*. These tokens are then assigned a POS tag that takes their context into consideration.

Given a set of POS tags P , I can generate a boolean mask for each story:

$$M_j = (f_{\text{mask}}(t_{j,1}), \dots, f_{\text{mask}}(t_{j,n})) \quad f_{\text{mask}}(t) = \begin{cases} 0 & \text{POS}(t) \in P \\ 1 & \text{POS}(t) \notin P \end{cases}$$

Since proper nouns and auxiliary verbs perform similar functions to nouns and verbs respectively, they are included in the noun and verb POS categories accordingly.

Attention Mask

To apply the POS masks during the generation of contextual embeddings for a word w , the mask values for w and its context C are first mapped to the GPT-2 tokens, yielding M'_w . They are then used as *attention masks*, indicating which tokens the model should attend to and which should not. The resulting contextual embedding is calculated by averaging only the unmasked hidden states:

$$\bar{e}_w = \frac{1}{L'} \sum_{i=1}^L m_w'^{(i)} h_8^{(i)} \quad L' = \sum_{i=1}^L m_w'^{(i)}$$

Rather than removing masked tokens from the input sequence — which would alter positional encodings and shift the attention context of remaining tokens — attention masking preserves the original sequence structure while selectively excluding the target POS category from the model’s attention computation.

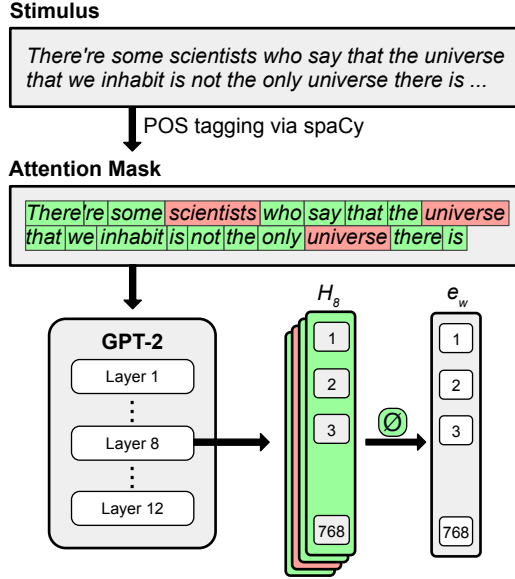


Figure 1. Contextual embedding pipeline with POS attention masking. A stimulus sentence is passed through a noun filter, which uses spaCy POS tags to generate an attention mask (green: unmasked; red: masked). The masked input is fed into GPT-2, whose hidden state at Layer 8, $H_8 \in \mathbb{R}^{L \times 768}$, is then averaged over unmasked tokens to produce a single contextualized embedding $e_w \in \mathbb{R}^{768}$ for the target word.

Results

The goal of this thesis was to determine whether the brain represents nouns and verbs in distinct areas, and if so, which of the three theoretical accounts — emergentist, combinatorial, or distributed — best characterises the observed pattern. To answer this question, I generated noun-masked and verb-masked contextual embeddings by applying attention masks to the input of GPT-2 during the generation of contextualized embeddings for the words in the 11 stories taken from *The Moth Radio Hour*. I then fit these features, in addition to acoustic, phonetic and [what] features to the BOLD responses of 9 subjects who listened to and read the stories, using Banded Ridge Regression. Finally, I compared the performance of the noun-masked and verb-masked model across the brain, to identify areas that were more sensitive to nouns and verbs respectively.

Baseline Performance

To ensure that the contextual embeddings were successfully able to predict brain activity, I first fit the unmasked contextual embeddings to the responses. After fitting the model, I calculated the per-voxel Pearson’s correlation r_v between the model predictions on the validation dataset and the actual responses. The resulting r_v values were then projected onto the cortical surface to yield a sensitivity map to the contextual embeddings. The contextual embeddings were able to predict brain activity in a large portion of the cortex, with the highest sensitivity in the temporal and frontal areas, which is consistent with previous studies (Schrimpf et al. 2021; Jain and A. Huth 2018). The average r_v across

the cortex was 0.21 for the listening modality and 0.20 for the reading modality, with some voxels reaching values as high as 0.65. This confirms the validity of the contextual embeddings as a feature space for predicting brain activity in response to naturalistic language stimuli.

POS-Masked Features

Before focusing on the noun-masked and verb-masked models, I first looked at the performance of a multitude of POS-masked features, including adjectives, adverbs, pronouns and interjections. For each POS-masked model, I calculated the drop in performance compared to the baseline unmasked model by subtracting the r_v values of the masked model from the r_v values of the baseline model. I then ranked the drop in performance for each POS category for each voxel individually and plotted the greatest drop on the flatmap in the respective color. This allowed me to identify which areas of the brain were more sensitive to each POS category (Fig 2a,b).

The results showed clear clusters of POS sensitivity across the brain with nouns being more prevalent in the Medial Parietal Cortex (MPC), while verbs were consistently present in the Temporo-Parietal Junction (TPJ). Nouns and verbs resulted in the greatest drop in performance, followed by pronouns and adverbs, while adjectives and interjections had the least impact on the model performance.

The distribution of POS tags in the training stories and the validation story was also analyzed, to verify that the validation story is a representative sample of the training data and that any differences in model accuracy between POS categories reflect the brain’s sensitivity to grammatical class rather than an artifact of unequal stimulus composition. (Fig 2c).

Noun and Verb Sensitivity

To investigate the sensitivity to nouns and verbs in more detail, I focused on the noun-masked and verb-masked models. For each voxel, I calculated the drop in performance compared to the baseline unmasked model for both the noun-masked and verb-masked models. I then plotted the respective drops in performance for each voxel on the flatmap using a bivariate color map, where the x-axis (orange) represents the drop in performance for the noun-masked model and the y-axis (blue) represents the drop in performance for the verb-masked model. This allowed me to identify areas of the brain that were more sensitive to nouns (points above the diagonal) and areas that were more sensitive to verbs (points below the diagonal), as well as areas that were sensitive to both (points on the diagonal) (Fig 3). The results showed a clear separation between noun-sensitive and verb-sensitive areas. Moreover, the greatest drops for both nouns and verbs were observed in high-level processing areas, such as the MPC, TPJ and frontal areas, while early sensory areas like the primary auditory cortex and the primary visual cortex were less affected by the masking of nouns and verbs.

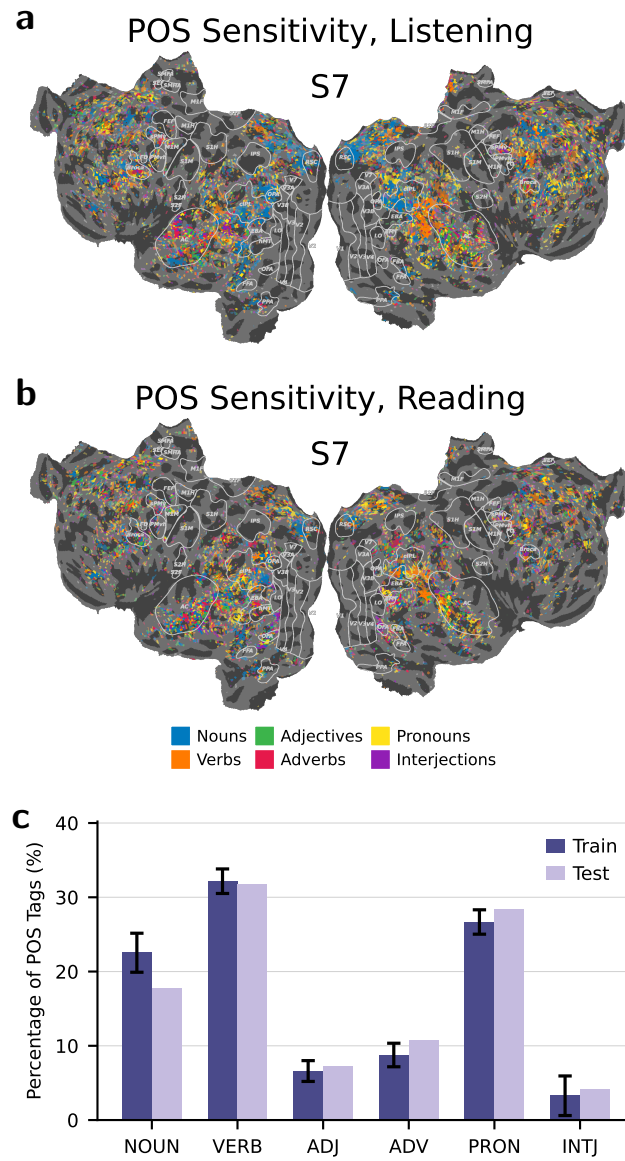


Figure 2. Parts of Speech sensitivity across the brain using all POS: nouns, verbs, adjectives, adverbs, pronouns and interjections. **a** Voxelwise model r values in the listening modality using POS-masked contextual embeddings were subtracted from the r values of the baseline contextual model to get the drop in accuracy. The drop in accuracy for each POS was then ranked for each voxel individually and the greatest drop was plotted on the flatmap in the respective color **b** Voxels are colored using the same procedure as in **a** but for the reading modality. **c** Distribution of POS tags in the training stories and the validation story. The percentage of each POS tag was calculated by counting the number of words with that tag and dividing it by the total number of words in the story. The average percentage across the 10 training stories is plotted on the left, while the percentage for the validation story is plotted on the right.

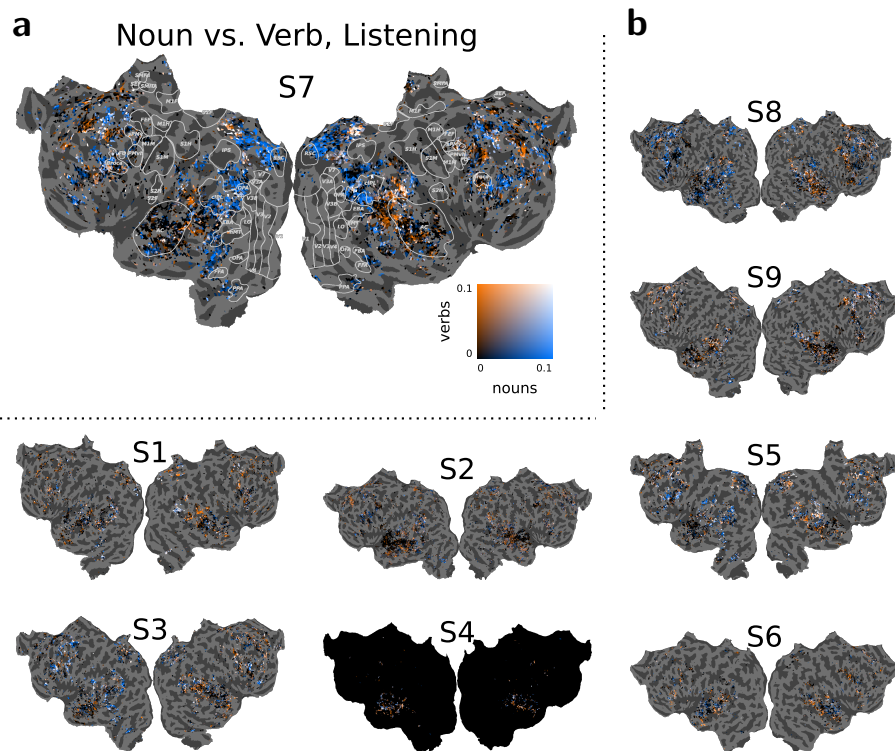


Figure 3. Noun and verb sensitivity across subjects in the listening modality. Voxelwise model r values in the listening modality for noun-masked and verb-masked contextual embeddings were subtracted from the r values of the baseline contextual model to get the drop in accuracy, then plotted using a bivariate color map. Blue points indicate a greater drop in performance for the noun-masked model while orange points indicate a greater drop for the verb-masked model. Grayscale points indicate a similar drop for both models. The large panel **a** shows the results for subject 07, while the smaller panels **b** show the results for the remaining subjects.

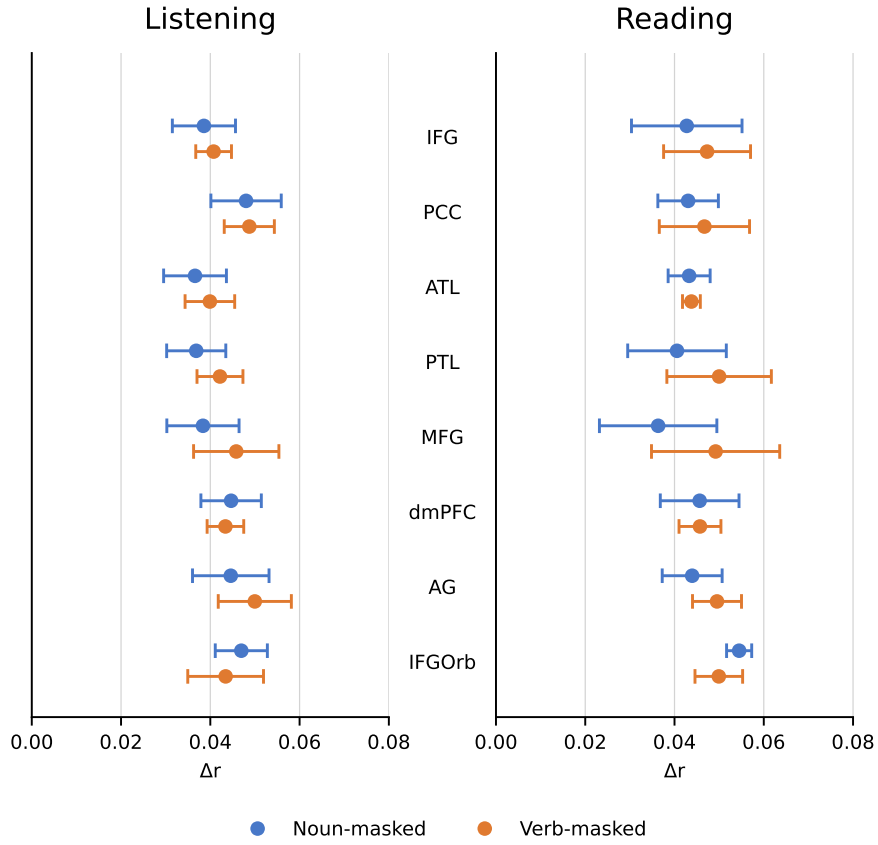


Figure 4. Noun and Verb sensitivity across Regions of Interest (ROIs). Points indicate the average drop in performance $\Delta r = r_{\text{baseline}} - r_{\text{masked}}$ for the noun-masked model (blue) and the verb-masked model (orange) across voxels within each ROI. Values are computed by first averaging the positive drops in performance within each ROI for each subject, then averaging across the cohort. Error bars represent ± 1 standard deviation across subjects.

Discussion

Table 1: Theoretical Accounts of Cortical Sensitivity to Nouns and Verbs

Theoretical Account	Key Evidence	L-IFG / Premotor	L-MTG / STG	L-Fusiform / ITG	L-Parietal	R-Hemisphere
Classic Fronto-Temporal (Category-specific loci)	Damasio & Tranel (1993) Cappa (2003)	V > N (Action/Motor)	N > V (Entities)	N > V	–	–
Semantic (Object-Action)	Mestres-Missé (2010) Bird et al. (2000)	V > N (Competition)	V > N (Action/Relational)	N > V (Concrete/Visual)	V > N (Tool use)	–
Combinatorial (Morphology/Inflection)	Tyler et al. (2004) Shapiro et al. (2005)	V > N (Verb inflection)	N = V (Shared stems)	–	–	N > V (Concrete Ns)
Distributed Account (Shared network)	Crepaldi et al. (2013) Caucheteux (2021)	N = V	N = V (Strict spatial segregation is rejected; highly interleaved network)	N = V	N = V	N = V

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Additional Figures

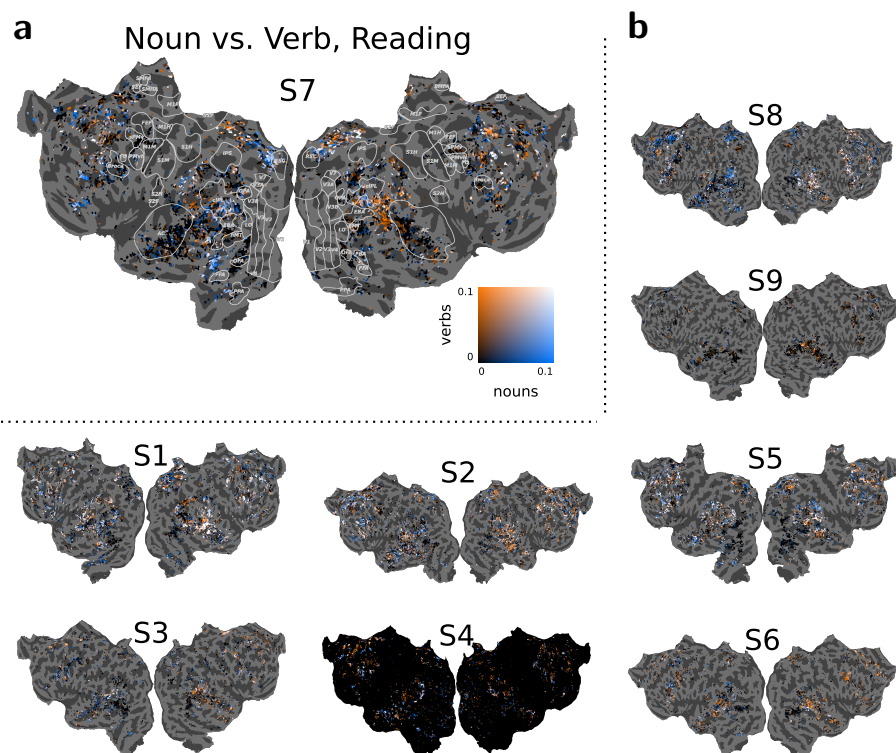


Figure 5. Noun and verb sensitivity across subjects. Each point represents